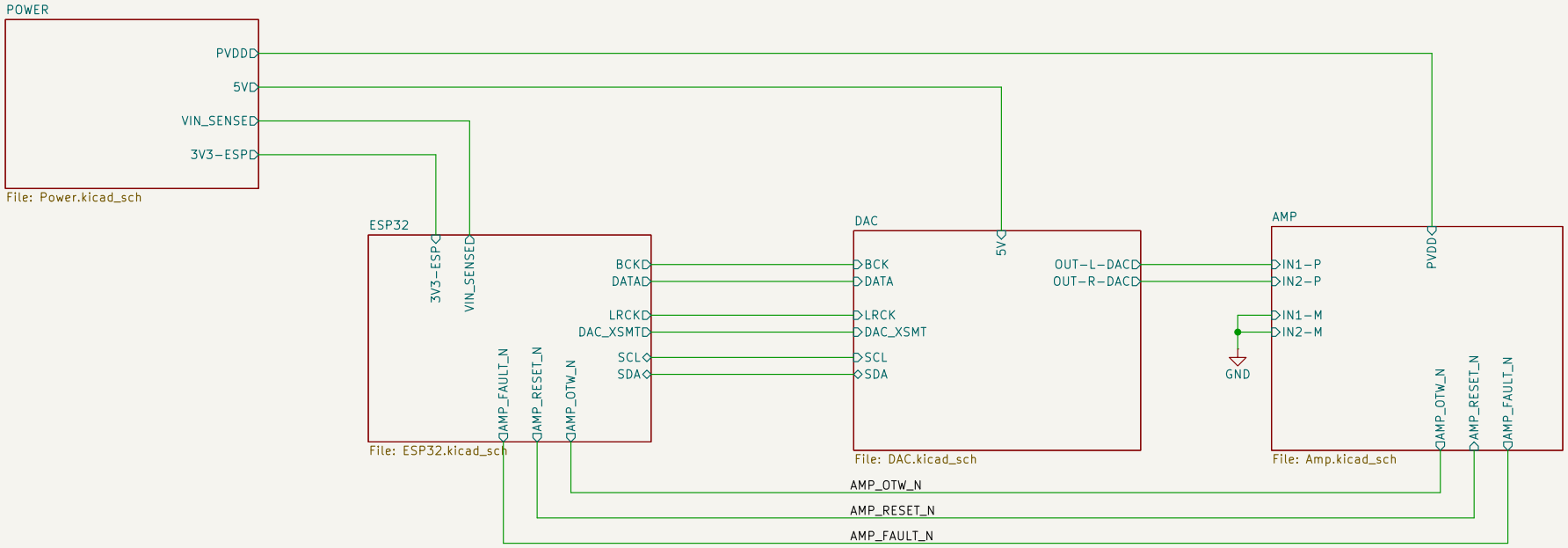
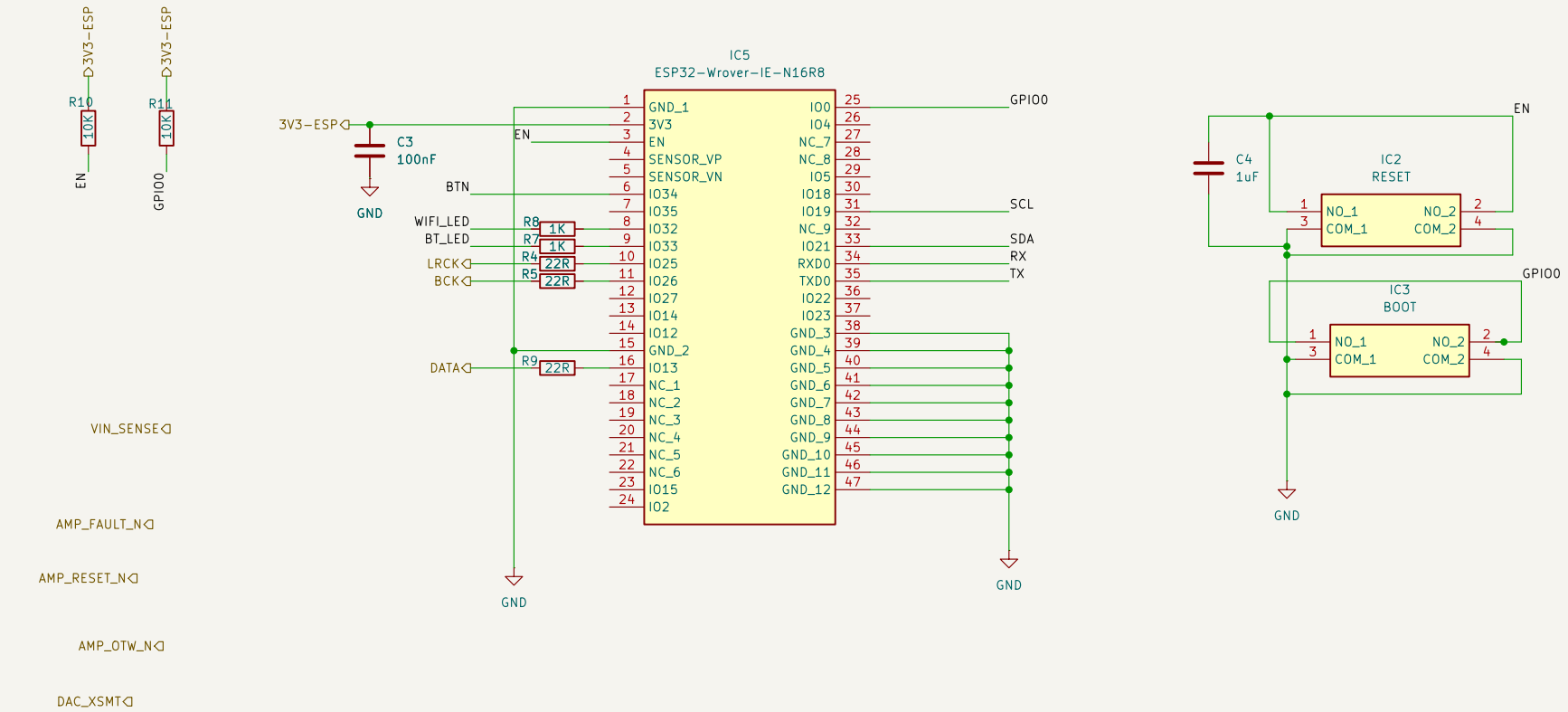


TPA3220 ACTIVE SPEAKER – ESP32 / PCM5122 / TPA3220

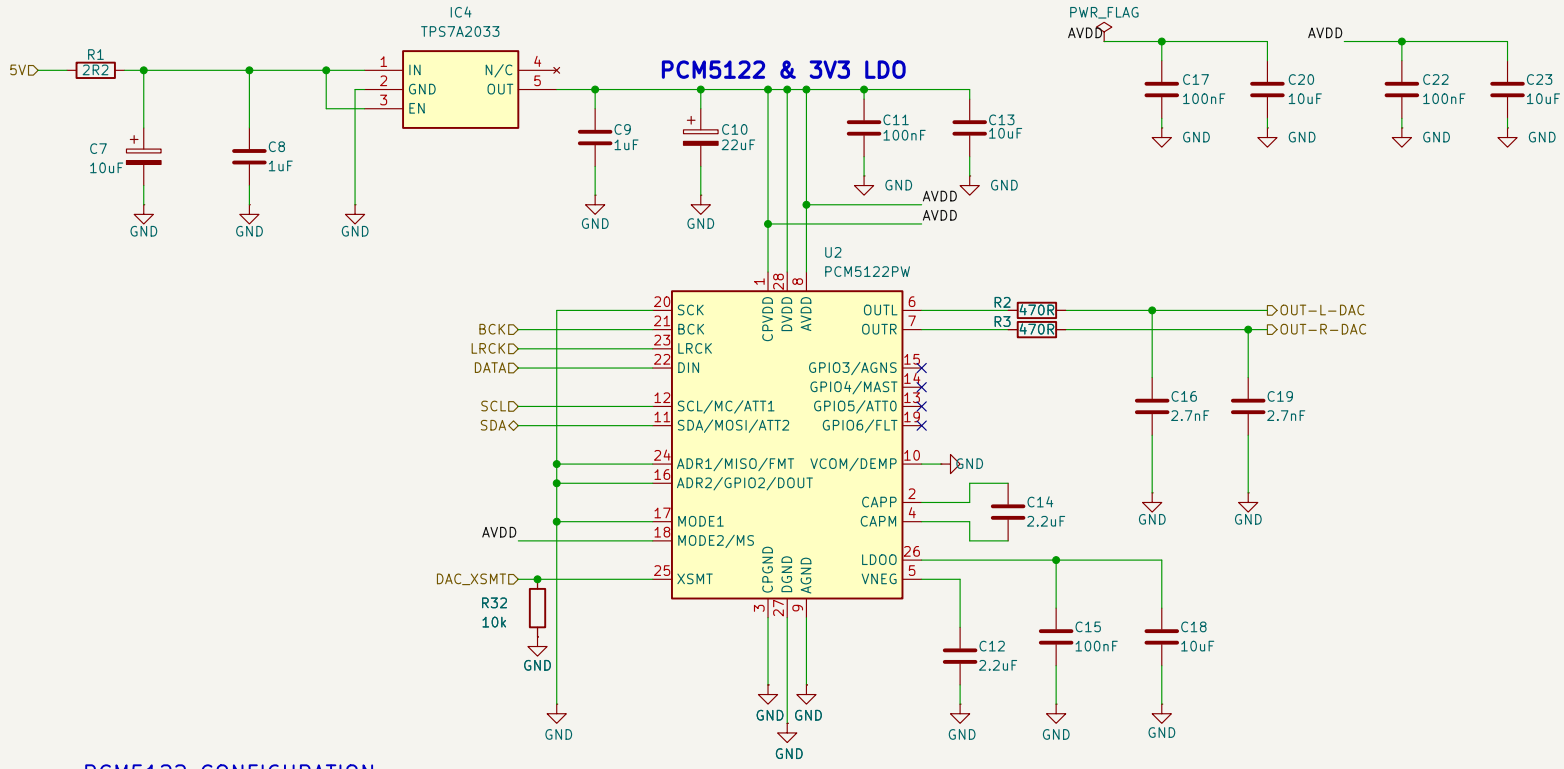
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

POWER TREE (12–16 V DC in, J1 on POWER sheet)
VIN –> F1 5 A T –> Q1 reverse-polarity P-FET –> D2 TVS –> PVDD
  PVDD –> TPA3220 PVDD + VDD (internal 5 V LDO for AVDD/GVDD) ... up to ~3.5 A peak
  PVDD –> U3 LMR51430XF buck 5.0 V / 500 kHz FPWM ..... ~0.35 A max
    5V –> U4 TLV1117LV33 –> 3V3-ESP (ESP32, pull-ups) ..... 0.15–0.25 A avg, 0.5 A peak
    5V –> R1 2R2 –> IC4 TPS7A2033 –> DAC AVDD/DVDD/CPVDD ..... ~40 mA
SIGNALS
  ESP32 IO25/26/13 –> PCM5122 LRCK/BCK/DIN (3-wire I2S, no MCLK, PLL from BCK)
  ESP32 IO21/22 <–> PCM5122 SDA/SCL, address 0x4C; PCM5122 does master volume
  PCM5122 OUTL –> TPA3220 CH1 –> J16 WOOFER, OUTR –> CH2 –> J17 TWEETER
  IO27 –> AMP_RESET_N, IO33 –> DAC_XSMT, IO34 <– AMP_FAULT_N, IO35 <– AMP_OTW_N,
  IO36 <– VIN_SENSE (PVDD/11)
POWER-UP: buck starts at PVDD ~9 V; ESP32 boots with amp held in RESET (4k7) and DAC soft-muted (XSMT 10k pull-down). Firmware: init PCM5122 over I2C, start I2S clocks, XSMT high, then RESET high (gain latched; ~20 ms ramp via 33 nF CMUTE).

Sheet: /		
File: TPA3126.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/5



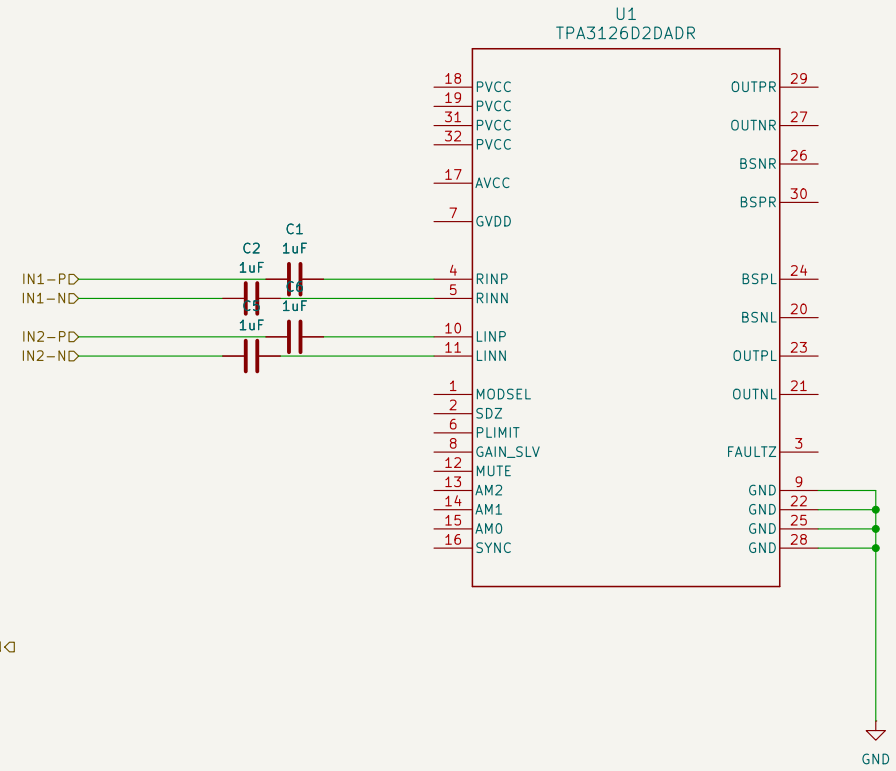
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File: ESP32.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/5



PCM5122 CONFIGURATION

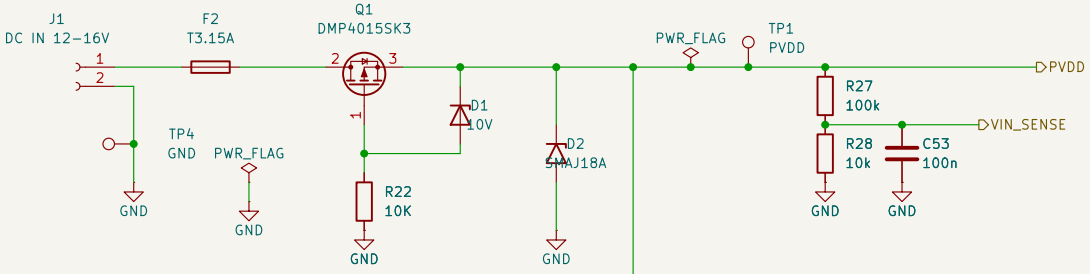
I2C mode: MODE1 = 0, MODE2 = 1. Address ADR2 = ADR1 = 0 -> 0x4C (7-bit).
SCK tied to GND: 3-wire I2S, PLL referenced to BCK. Firmware must set
P0-R13 SRCREF = BCK (0x10) and use BCK = 64 fs (32-bit slots) or 32 fs.
Volume: P0-R61/R62 digital volume over I2C (master volume).
OUTL -> amp CH1 (woofer). OUTR -> amp CH2 (tweeter); 2.1 Vrms FS,
ground-centred. 470R + 2.7 nF output filter = 125 kHz (TI: 470R/2.2 nF).
XSMT: 10k pull-down -> soft-muted until ESP32 drives DAC_XSMT high.
Assert XSMT low >= 150/fs + 0.2 ms before power is removed (TI 10.2.3).
VCOM (pin 10) to GND for default VREF mode. GPIO3-6 internal pull-downs.
DVDD = AVDD = CPVDD = 3.3 V from IC4 (TPS7A2033); LDOO = 1.8 V core.

Sheet: /DAC/	
File: DAC.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 10.0.5	Rev: Id: 3/5



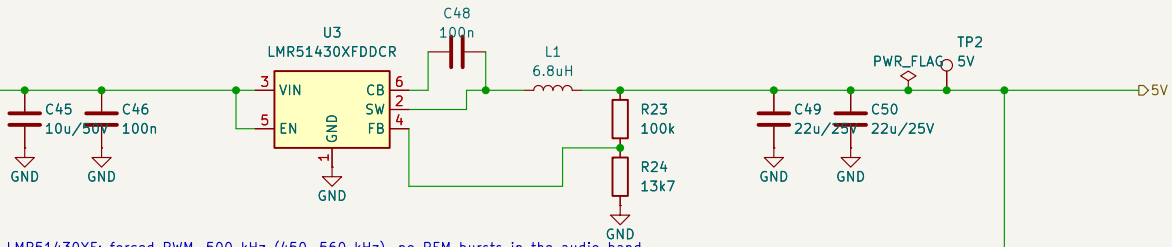
Sheet: /AMP/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 4/5

INPUT PROTECTION



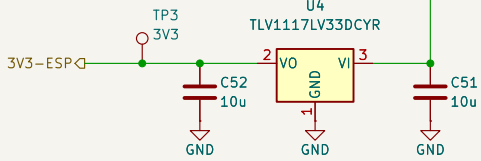
J1: 12–16 V DC from panel-mounted jack (pin 1 = +, pin 2 = GND).
F1 5 A time-lag. Q1 high-side P-FET: blocks reverse polarity; body diode starts it, then VGS = −VIN via R22. D1 clamps VGS (DMP4015SK3 VGS max +/−25 V).
D2 SMAJ18A: VWM 18 V, VC 29.2 V max – below TPA3220 PVDD abs max (37 V), LMR51430 VIN abs max (38 V) and the 35 V bulk electrolytics.
PVDD/11 –> VIN_SENSE (ESP32 GPIO36) for power-fail detection.

5 V BUCK (feeds ESP32 LDO + DAC LDO)



LMR51430XF: forced PWM, 500 kHz (450–560 kHz), no PFM bursts in the audio band.
Vout = 0.6 V x (1 + 100k/13.7k) = 4.98 V. EN tied to VIN (internal UVLO 4.5 V).
Firmware must assert DAC_XSMT low then AMP_RESET_N low when VIN_SENSE falls (~9.5 V), before TPA3220 UVP (6.4 V). L1 ripple ~1.0 A p-p at 16 V in; peak IL ~1.1 A.

3V3 LDO (ESP32 only)



TLV1117LV33: 1 A, Vdo 0.455 V @ 1 A.
Pdiss ~0.3 W avg (180 mA), 0.85 W bursts.
SOT-223 tab to copper (RthJA 63 C/W).
ESP32 22 uF + 100 nF on ESP32 sheet.

Sheet: /POWER/		
File: Power.kicad_sch		
Title: Power input, protection and regulation		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 5/5